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Dip. in Engg. (Fifth Semester)/ P. T. D. C. (Seventh Semester) EXAMINATION, May-June, 2015

(Branch : Elect.)

POWER ELECTRONICS

Time : Three Hours] [*Maximum Marks : 100*

[*Minimum Pass Marks : 35*

Note : All questions are compulsory, unless mentioned otherwise.

1. (a) With the neat block diagram, explain the functional element of power electronic system. 5

   (b) Explain free-wheeling action of a diode. 5

2. Show that string efficiency for two parallel converted SCR is usually less than one. Discuss the problem associated with the parallel operation of SCR's and how these are overcome. 10

Or

Describe the working of a 3-phase F. W. in the rectifier made with RLE load. Discuss how one pair of SCR's is commutated by an incoming pair of SCR's. Illustrate

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your answer with waveform for source voltage E, output voltage and current, source current, current through and voltage across one thyristor. Derive an expression for V_0 and I_0.

3. (a) Explain with the help of neat power diagram and associated waveform, the operation a branch connection of transform. 5

   (b) Explain the difference between a Triac and an SCR. 5

4. With the help of neat circuit diagram and waveform, explain briefly the operation of transistorised three-phase bridge inverter with resistive load (1) 120° conduction made. 10

Or

Explain the working of the following :

(a) CSI with resistive load

(b) Parallel inverter

5. Draw the schematics of step-down and step-up chopper and derive an expression for output voltage in terms of duty cycle for step-down and step-up chopper. 10

Or

With the help of a circuit diagram, explain the working of class D chopper.

6. Explain the basic principle of operation of a cyclo-converter with a neat equivalent circuit diagram. Also draw and discuss the waveform illustrating the operation of the idealized cycloconverter circuit, with load of various displacement angle. 10

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Or

With the help of block diagram explain the basic principle of operation of SMPS.

7. Explain with the block diagram, the basic type of voltage regulator circuit. 10

Or

Draw and explain block diagram of IC regulator.

8. What is regenerative breaking ? Describe the regenerative breaking of a chopper fed separately excited d. c. motor. Illustrate your answer with circuit diagram and relevant waveform. 10

Or

'Induction motor speed control with constant supply voltage and reduced supply frequency is rarely used in practice.' Justify this statement.

9. Draw the functional diagram of 8085 by showing its main component. 10

10. Write short notes on any *two* of the following : 5 each

(a) PLC

(b) Micro-controller

(c) Micro-computer

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